

Department of Electrical, Electronic, and Computer Engineering

2019EBN111E01
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Module EBN111
Electricity and Electronics
 June 2019



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
 Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

Exam Information:			
Maximum marks:	92	Full marks:	90
Duration of exam:	200 minutes	Open/closed book	Closed
Total number of pages (including this page):		44 + OCR	
IMPORTANT			
1. <i>The examination regulations of the University of Pretoria apply.</i>			
2. <i>All questions must be answered on the Optical Character Recognition (OCR) form.</i>			
3. <i>Carefully read and follow the instructions on the OCR form.</i>			
4. <i>Answers must include units when applicable!</i>			
5. <i>Final answers must be written as real numbers, and not as fractions.</i>			
6. <i>Round off answers to 3 significant digits when applicable.</i>			
Lecturers	1. D Ramotsoela		
	2. X. Ye		
Internal Moderator	F. Paluncic		

INSTRUCTIONS	
Do step 0 in the first 5 minutes of the session. (-5 min to 0 min) Do step 1 in the first 180 minutes of the exam. (0 min to 180 min) Do step 2 within the following 20 minutes. (180 min to 200 min)	
STEP 0 (-5 min to 0 min): Complete the front page of the OCR form. Also write the assessment ID and student number on ALL pages of the OCR form.	
STEP 1 (0 min to 180 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units when applicable. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.	
STEP 2 (180 min to 200 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form. - The Assessment ID is: 2019EBN111E01 - Use a mechanical pencil with 0.5mm HB lead to complete the OCR form. - Each cell must only have one character. - Characters must not touch or cross the cell boundaries. - Do not use commas! Use full stops.	

Examples

01	6	8	.	4					°									
02	3	4	0	+	j	2	0		Ω									
03	1	5	.	2	∠	4	2		°	V								
04	1	7								r	a	d	÷	s				

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SECTION 1 [46]

Questions 1–2 [4]	
01	A current of 2 A enters the negative terminal of an element between 0.5 s and 1 s. Determine the energy associated with the element in this period if the terminal voltage is $v(t) = 5(t - e^{-0.5t})$ V.
02	A 6 kW oven is able to roast 4 chickens in 120 minutes. A restaurant uses this oven to prepare 16 chickens every day. If electricity costs R 3/kWh, how much does the restaurant spend on electricity (per month of 30 days)?

01:

$$\begin{aligned} E &= \int_{t_1}^{t_2} -v(t)i(t)dt \\ &= \int_{0.5}^1 -5(t - e^{-0.5t})(2)dt \\ &= -305 \text{ mJ.} \end{aligned}$$

02:

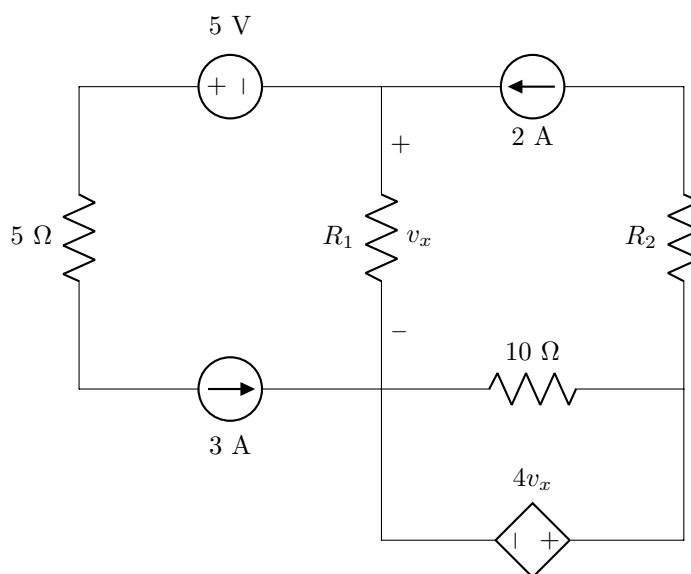
$$\begin{aligned} \text{Time} &= \left[16 \text{ chickens/day} \times \frac{120 \text{ min}}{4 \text{ chickens}} \right] (30 \text{ day/month}) \\ &= 14400 \text{ min/month} \\ &= 240 \text{ hrs/month.} \\ \text{Energy} &= (240 \text{ hrs/month} \times 6 \text{ kW}) \\ &= 1440 \text{ kWh/month.} \\ \text{Monthly Cost} &= \text{R } 3/\text{kWh} \times 1440 \text{ kWh} \\ &= \text{R } 4320. \end{aligned}$$

Questions 3–4 [4]

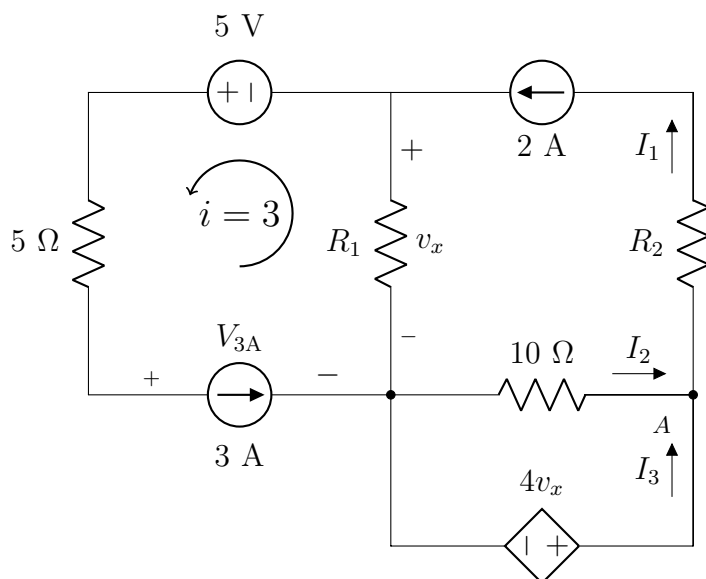
Assume that $v_x = 5$ V,

03 Find the power associated with the 3 A current source (P_I).

04 Find the power associated with the dependent voltage source (P_V).



Solution:



03:
KVL at left loop:

$$\begin{aligned}
 -5 + 5(3) + V_{3A} - 5 &= 0 \\
 V_{3A} &= -5 \text{ V.} \\
 P_I &= VI \\
 &= 5(-3) = -15 \text{ W.}
 \end{aligned}$$

04:

KCL at node A:

$$I_3 + I_2 = I_1$$

$$I_3 + \frac{-4(5)}{10} = 2$$

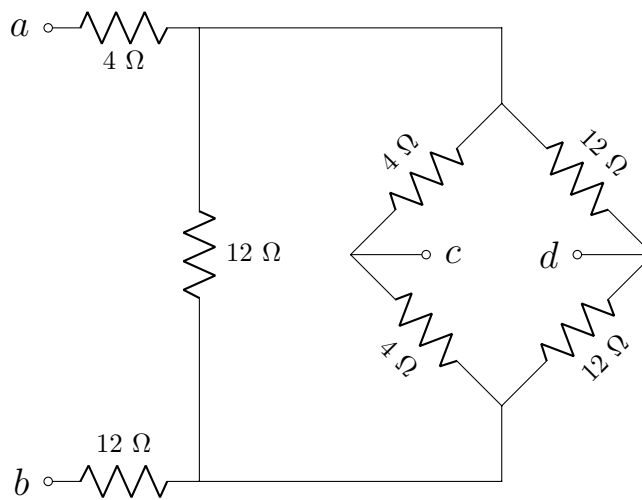
$$I_3 = 4 \text{ A.}$$

$$\begin{aligned} P_V &= -VI \\ &= -20(4) \\ &= -80 \text{ W.} \end{aligned}$$

Questions 5–6 [4]

05 Find the equivalent resistance w.r.t. terminals $a - b$ (R_{ab}).

06 Find the equivalent resistance w.r.t. terminals $c - d$ (R_{cd}).

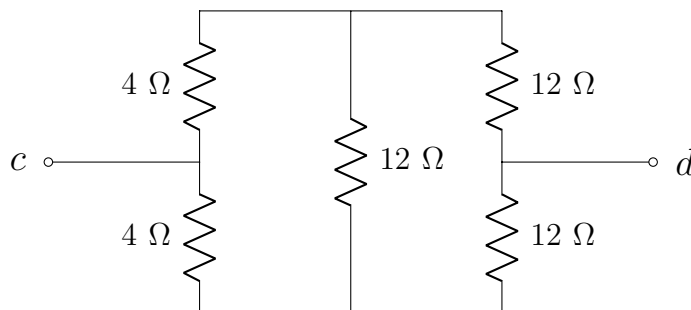


05:

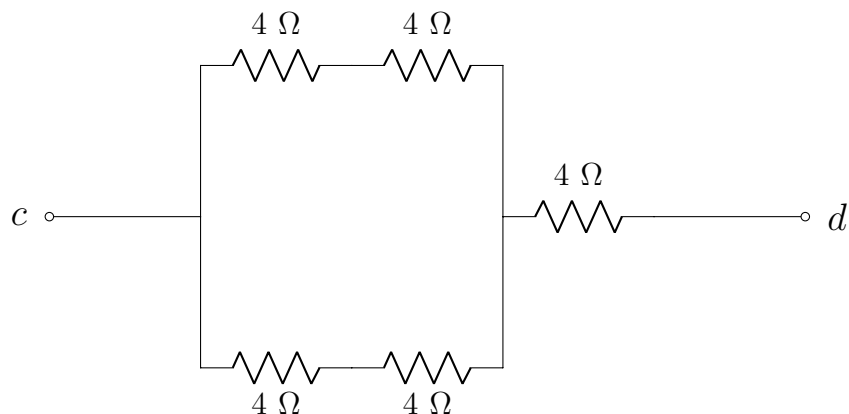
$$\begin{aligned} R_{ab} &= [12 // [(4 + 4) // (12 + 12)]] + 4 + 12 \\ &= [12 // 8 // 24] + 16 \\ &= 20 \, \Omega. \end{aligned}$$

06:

The circuit can be rearranged as follows (disregarding open terminals a and b and their respective resistors).



A delta to wye transformation can be applied on the right hand delta network to yield the circuit below.

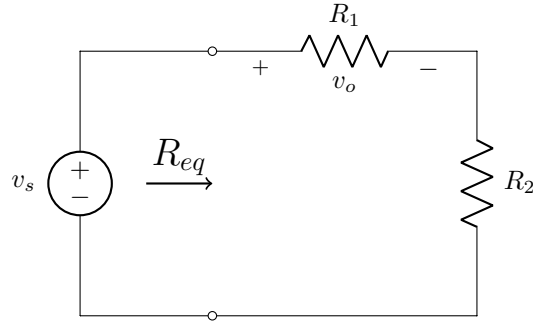


$$R_{cd} = 8 \text{ } \Omega.$$

Question 7 [2]

07

Find R_2 if $\frac{v_o}{v_s} = 0.05$ and $R_{eq} = 20 \text{ k}\Omega$.



07:

$$v_o = v_s \left(\frac{R_1}{R_1 + R_2} \right)$$

$$\frac{v_o}{v_s} = 0.05 = \frac{R_1}{R_1 + R_2}, \quad R_1 + R_2 = 20 \text{ k}\Omega$$

$$0.005 = \frac{R_1}{20}$$

$$R_1 = 1 \text{ k}\Omega$$

$$R_2 = 19 \text{ k}\Omega.$$

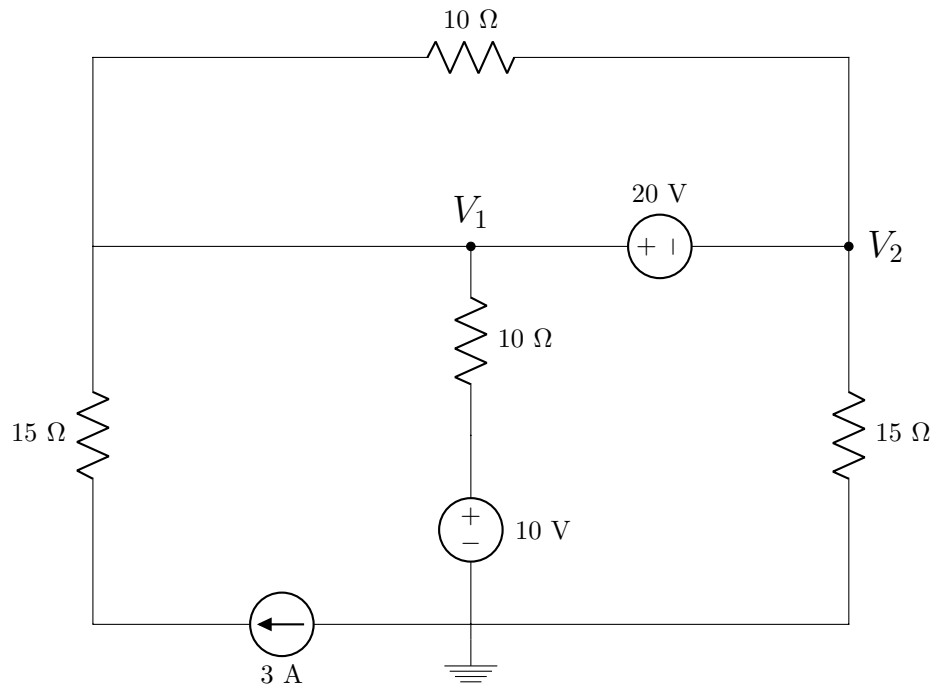
Questions 8–11 [4]

Use nodal analysis to set up two equations of the following form and find the values a , b , c , and d .

$$aV_1 + bV_2 = 20$$

$$cV_1 + dV_2 = 120$$

08	a	09	b
10	c	11	d



08-11: Supernode:

$$V_1 - 20 = V_2$$

$$V_1 - V_2 = 20.$$

$$\begin{aligned} \frac{V_1 - 10}{10} - 3 + \frac{V_2}{15} &= 0 \\ 3(V_1 - 10) - 90 + 2V_2 &= 0 \\ 3V_1 - 30 - 90 + 2V_2 &= 0 \\ 3V_1 + 2V_2 &= 120. \end{aligned}$$

Thus, $a = 1$, $b = -1$, $c = 3$, and $d = 2$.

Questions 12–15 [4]

Use mesh analysis to set up two equations of the following form and find the values e , f , g , and h .

$$ei_1 + fi_2 = 9$$

$$15i_1 + gi_2 = h$$

12

e

13

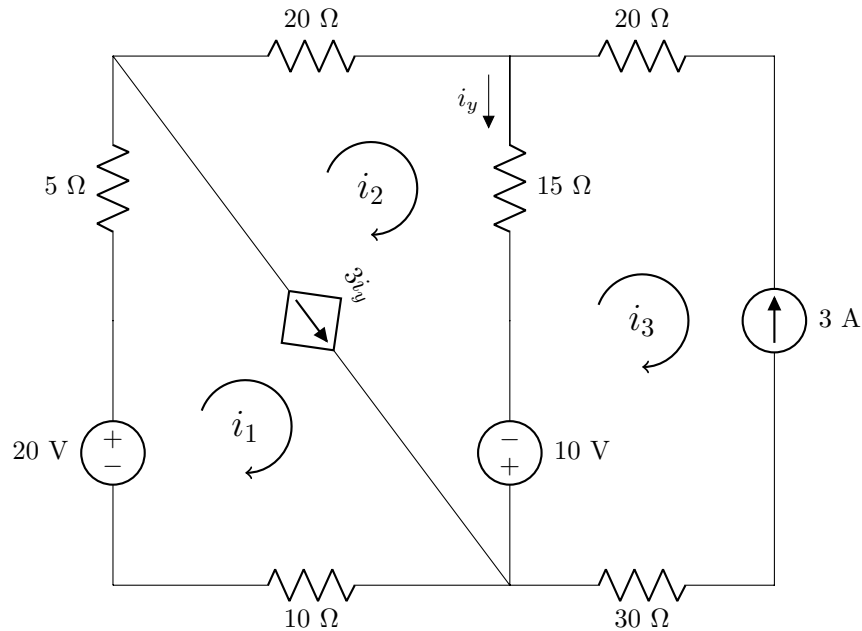
f

14

g

15

h



12-15:

$$i_y = i_2 + 3$$

Loop of i_1 and i_2 :

$$10i_1 - 20 + 5i_1 + 20i_2 + 15(i_2 + 3) = 10$$

$$15i_1 + 35i_2 = -15.$$

Dependent current source:

$$i_1 - 3i_y = i_2$$

$$i_1 - 3(i_2 + 3) = i_2$$

$$i_1 - 3i_2 - 9 = i_2$$

$$i_1 - 4i_2 = 9.$$

Thus, $e = 1$, $f = -4$, $g = 35$, and $h = -15$.

Questions 16–17 [2]

Use inspection to set up a matrix equation of the following form and find the values i and j .

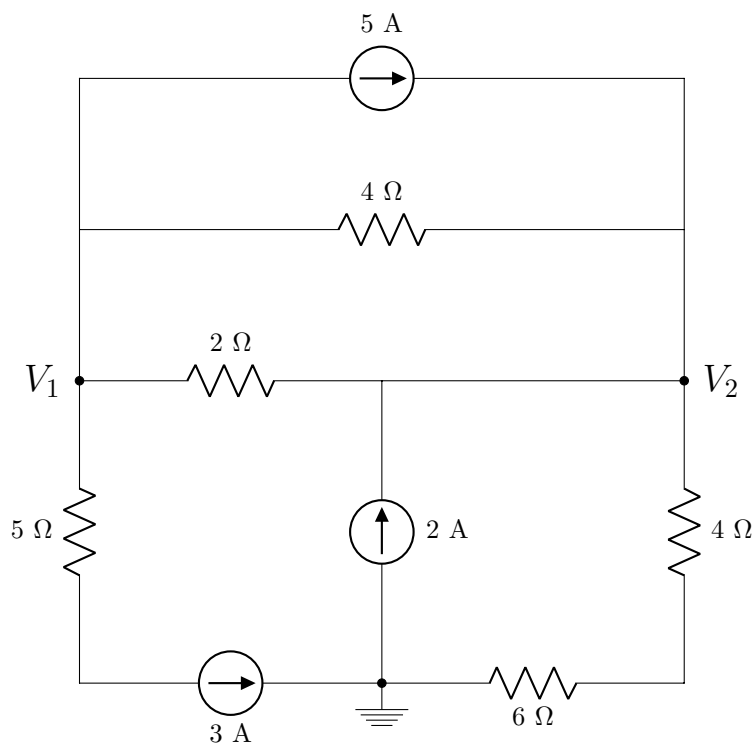
$$\begin{bmatrix} 0.75 & -0.75 \\ -0.75 & j \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} i \\ 7 \end{bmatrix}$$

16

i

17

j



16-17:

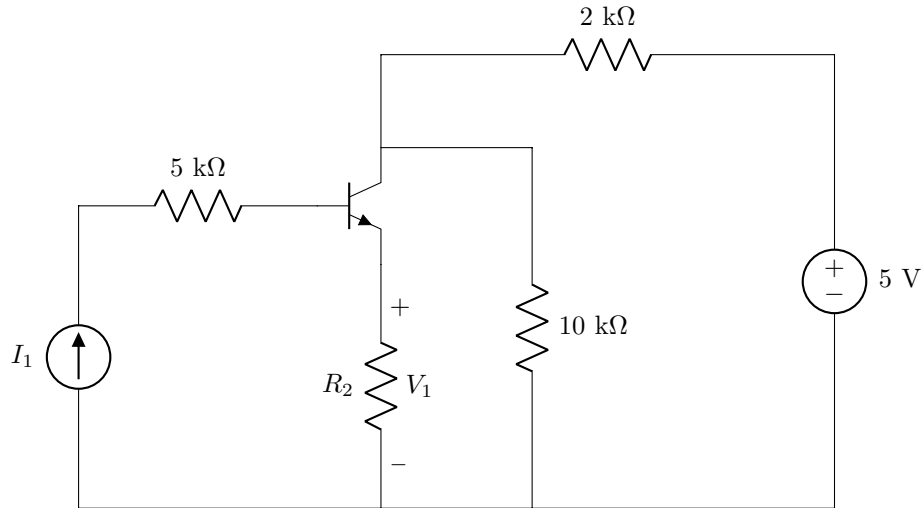
$$i = -5 - 3 = -8.$$

$$j = \frac{1}{4} + \frac{1}{2} + \frac{1}{10} = 0.85.$$

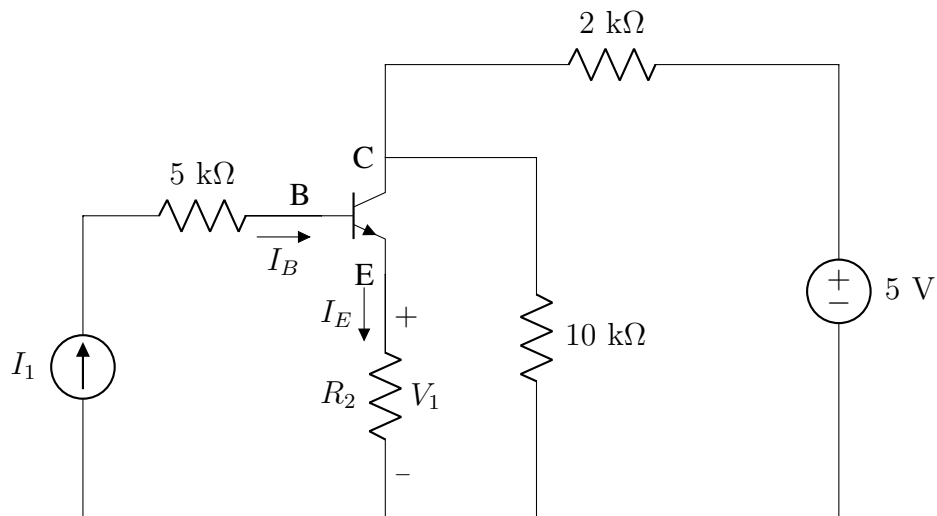
Questions 18–19 [4]

In the circuit below, $\beta = 99$, and $V_{BE} = 0.7$ V,

- | | |
|-----------|---|
| 18 | Find I_1 if $V_1 = 3$ V and $R_2 = 2$ k Ω . |
| 19 | Find V_{CE} if $R_2 = 0$ Ω and $I_1 = 12$ μ A. |



18: Using relationship between emitter and base currents:



$$\begin{aligned}
 I_E &= \frac{V_1}{R_2} = \frac{3}{2000} = 1.5 \text{ mA} = (\beta + 1)I_B \\
 &= 100I_B \\
 I_B &= I_1 = 15 \text{ } \mu\text{A}.
 \end{aligned}$$

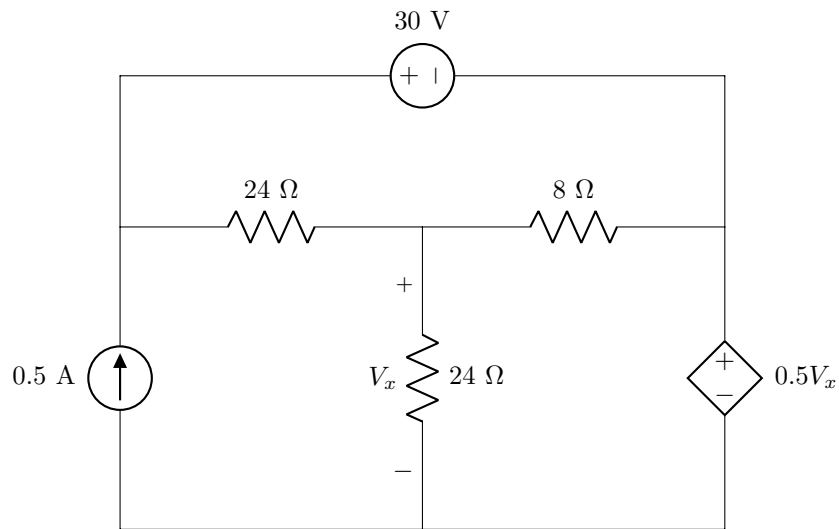
19: Use KCL at collector node C:

$$\begin{aligned}I_C + \frac{V_{CE}}{10000} + \frac{V_{CE} - 5}{2000} &= 0 \\99I_B + \frac{V_{CE}}{10000} + \frac{V_{CE} - 5}{2000} &= 0 \\99(0.012 \text{ mA}) + \frac{V_{CE}}{10000} + \frac{V_{CE} - 5}{2000} &= 0 \\V_{CE} &= 2.19 \text{ V.}\end{aligned}$$

Questions 20–21 [4]

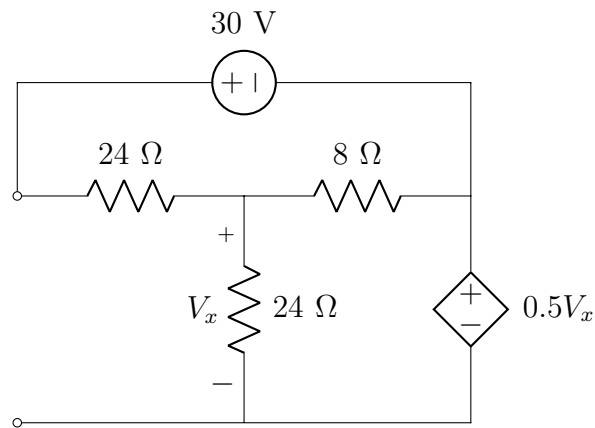
20 Find V_x^V , the contribution of the independent voltage source to V_x .

21 Find V_x^I , the contribution of the current source to V_x .



20:

Turning off current source results in the circuit below.

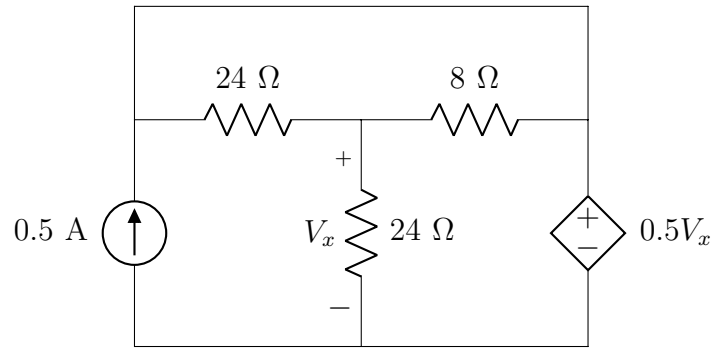


$$\frac{V_x - 30 - 0.5V_x}{24} + \frac{V_x}{24} + \frac{V_x - 0.5V_x}{8} = 0$$

$$V_x = 10 \text{ V.}$$

21:

Turning off voltage source results in the circuit below.



$$\frac{V_x - 0.5V_x}{24} + \frac{V_x - 0.5V_x}{8} + \frac{V_x}{24} = 0$$

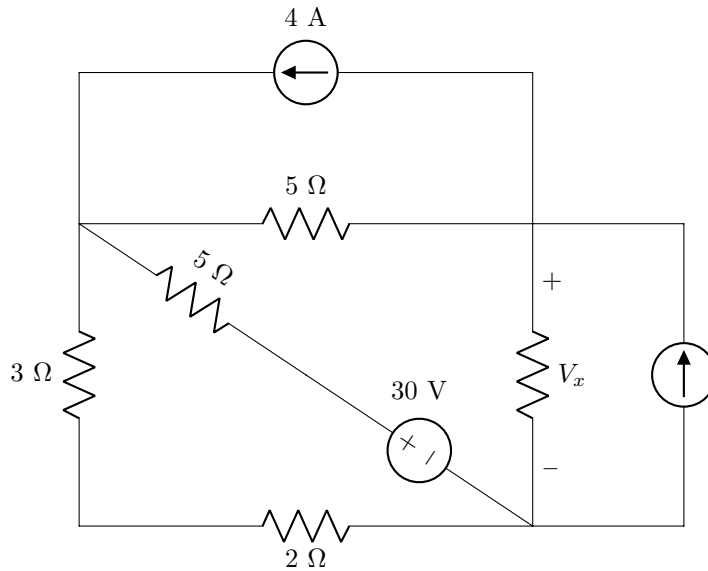
$$V_x = 0 \text{ V.}$$

Questions 22–23 [4]

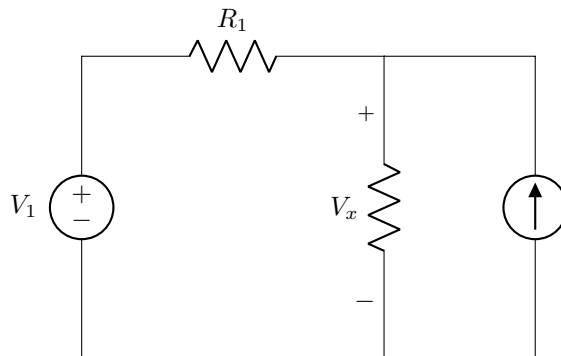
Use source transformation to determine V_1 and R_1 in *circuit B* such that it is equivalent to *circuit A*.

22	V_1
23	R_1

Circuit A:



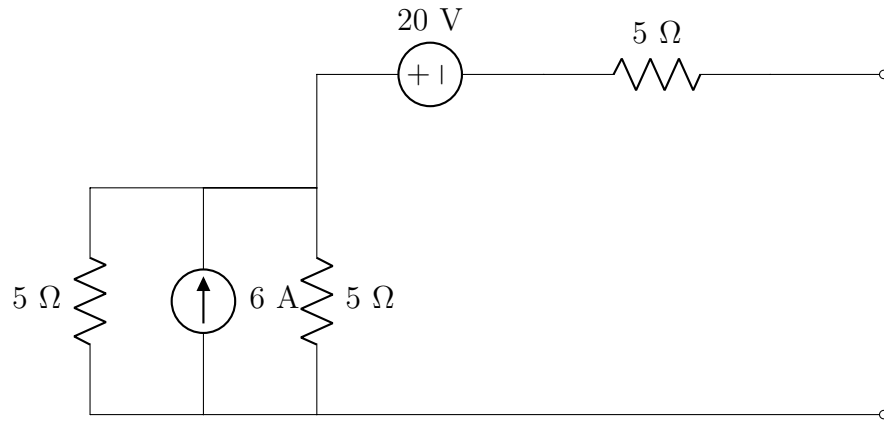
Circuit B:



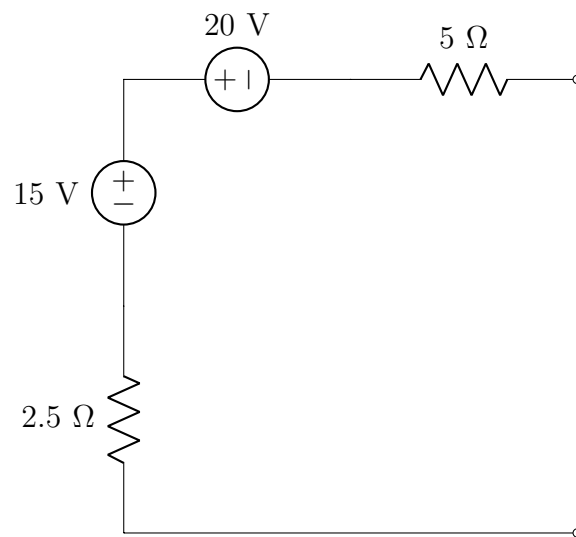
Turn over for additional space for Questions 22–23.

Additional space for Questions 22–23.

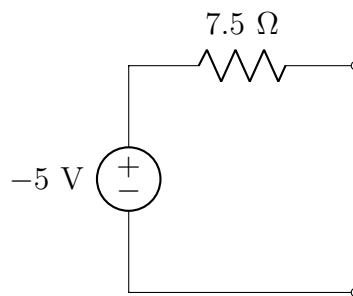
Step 1:



Step 2:



Step 3:



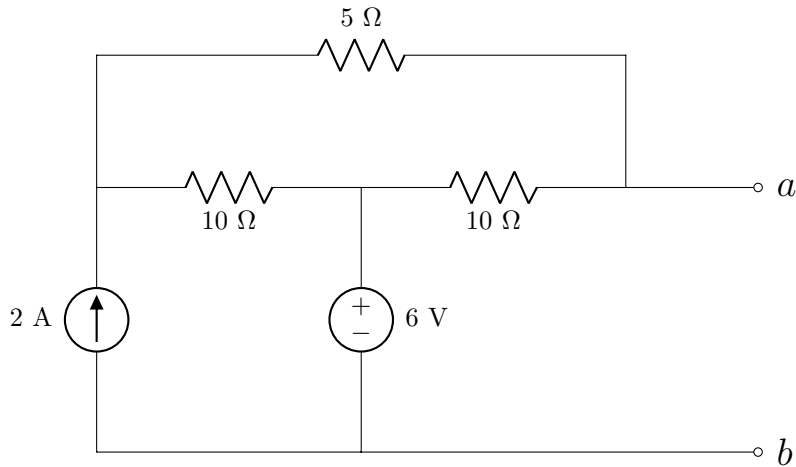
Thus, $R_1 = 7.5\ \Omega$, and $V_1 = -5\text{ V}$.

Questions 24–25 [4]

Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} w.r.t. terminals $a - b$.

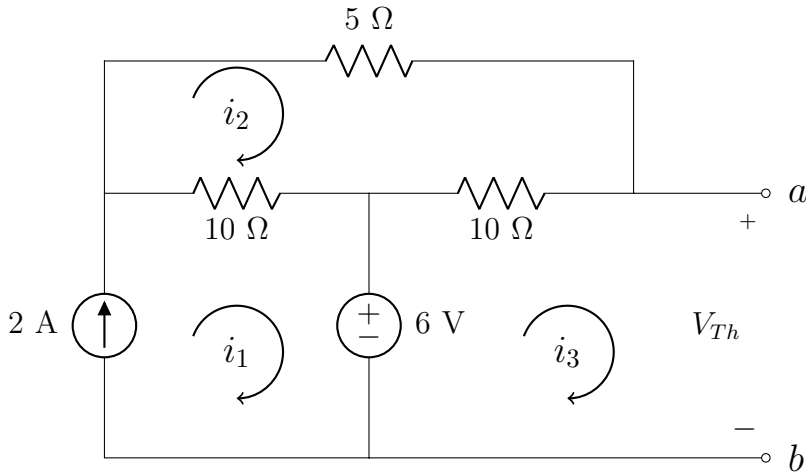
24 V_{Th}

25 R_{Th}



24:

The open circuit voltage between terminals a and b can be determined by using mesh analysis.



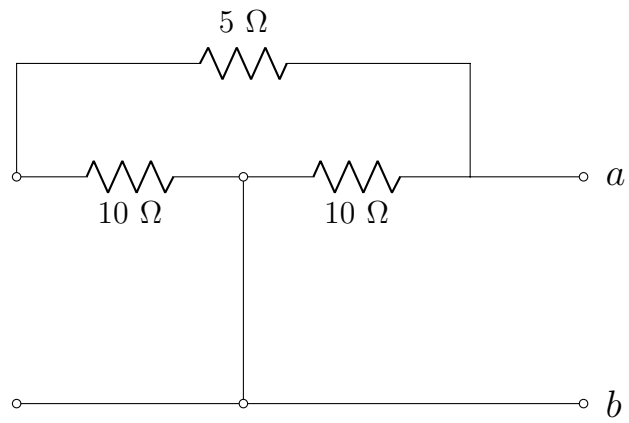
KVL at loop i_2 ($i_1 = 2$ A):

$$\begin{aligned} 25i_2 - 10i_1 &= 0 \\ 25i_2 &= 20 \\ i_2 &= 0.8 \text{ A.} \end{aligned}$$

KVL at loop i_3 :

$$\begin{aligned} -6 + 10(i_2) + V_{Th} &= 0 \\ V_{Th} &= 14 \text{ V.} \end{aligned}$$

25:

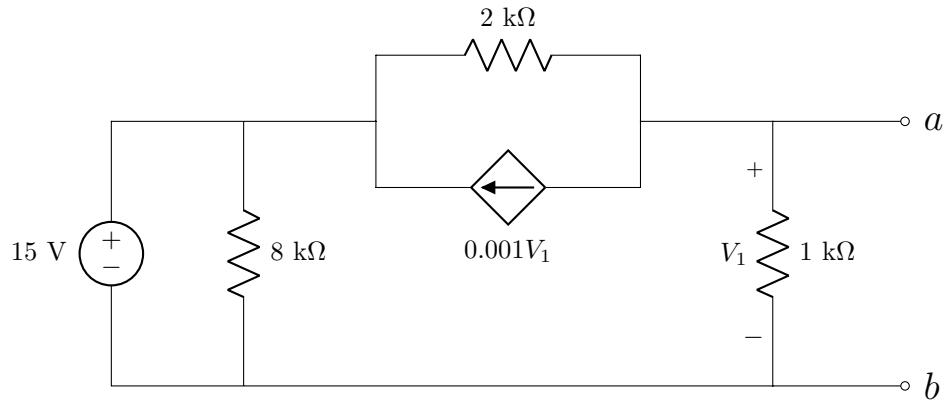


$$\begin{aligned} R_{Th} &= 15//10 \\ &= 6\ \Omega. \end{aligned}$$

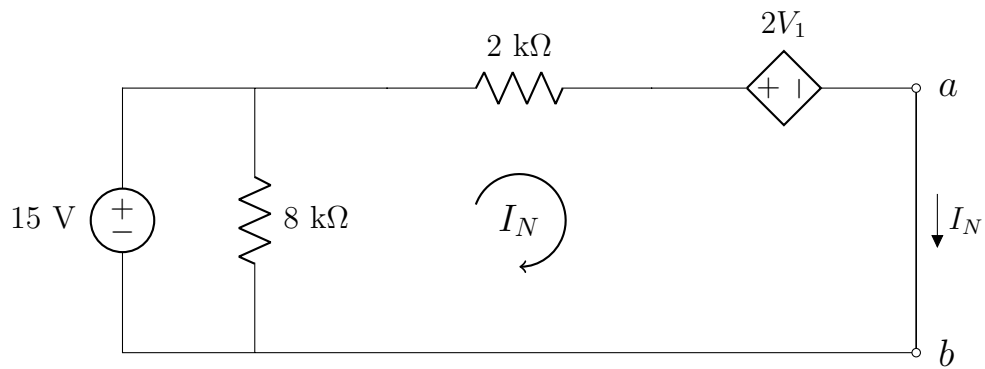
Questions 26–27 [4]

Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals $a - b$.

26	I_N
27	R_N



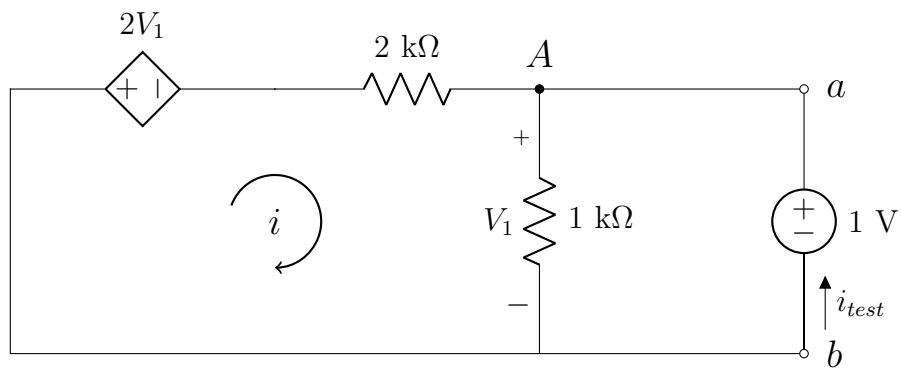
26:



Using KVL:

$$\begin{aligned}
 -15 + 2000I_N + 0.001(0) &= 0 \\
 I_N &= 7.5 \text{ mA.}
 \end{aligned}$$

27:



Using KVL for loop i :

$$\begin{aligned}2 + 2000i + 1 &= 0 \\ i &= -1.5 \text{ mA}.\end{aligned}$$

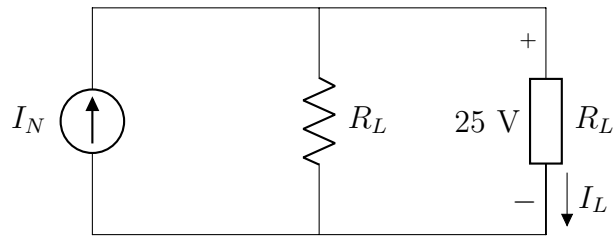
Using KCL at node A:

$$\begin{aligned}i_{test} &= i_{V_1} - i \\ &= 1 \text{ mA} + 1.5 \text{ mA} \\ i_{test} &= 2.5 \text{ mA}.\end{aligned}$$

$$R_N = \frac{1 \text{ V}}{0.025 \text{ A}} = 400 \text{ } \Omega.$$

Question 28 [2]**28**

A Norton equivalent circuit is connected to a load resistor R_L , and the maximum power P_{max} is delivered to the load. If the voltage across R_L is 25 V and $P_{max} = 800$ mW, find the Norton equivalent current I_N .

28:

For maximum power transfer, $R_N = R_L$.

$$P_{max} = 800 \text{ mW} = VI_L$$

$$0.8 = 25I_L$$

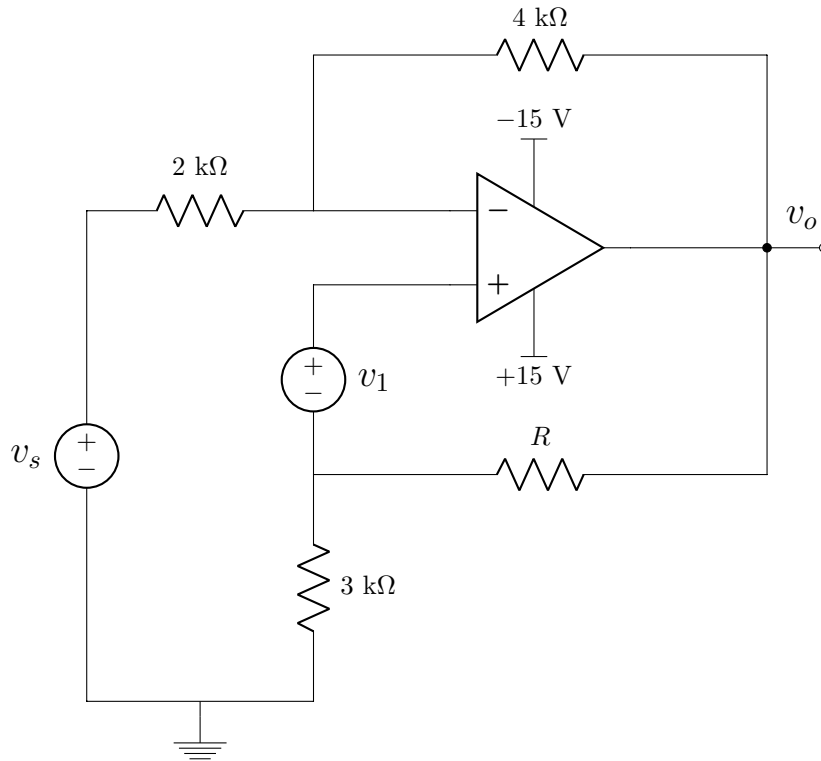
$$I_L = 32 \text{ mA.}$$

$$I_N = 2I_L = 64 \text{ mA.}$$

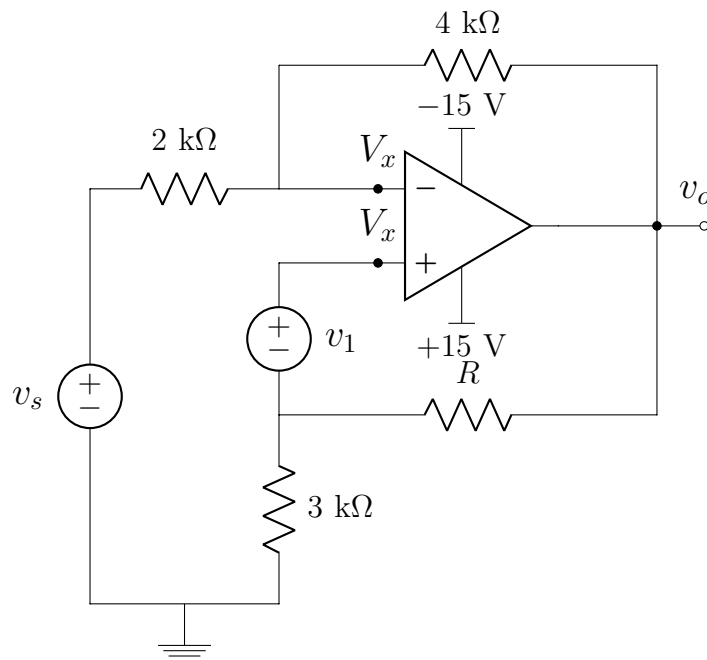
SECTION 2 [46]

Questions 29–30 [4]

- | | |
|-----------|--|
| 29 | Find v_s if $v_1 = 2\text{ V}$, $R = 5\text{ k}\Omega$, and $v_o = -4\text{ V}$. |
| 30 | If $v_1 = 0$ and $R \rightarrow \infty$, find the maximum value of v_s before the op amp enters saturation. |



29:



Using voltage division:

$$\begin{aligned}v_x &= -4 \left(\frac{3}{5+3} \right) + 2 \\&= 0.5 \text{ V.}\end{aligned}$$

Using KCL at V_x^- :

$$\begin{aligned}\frac{0.5 - v_s}{2000} + \frac{0.5 + 4}{4000} &= 0 \\v_s &= 2.75 \text{ V.}\end{aligned}$$

30:

The circuit becomes an inverting amplifier:

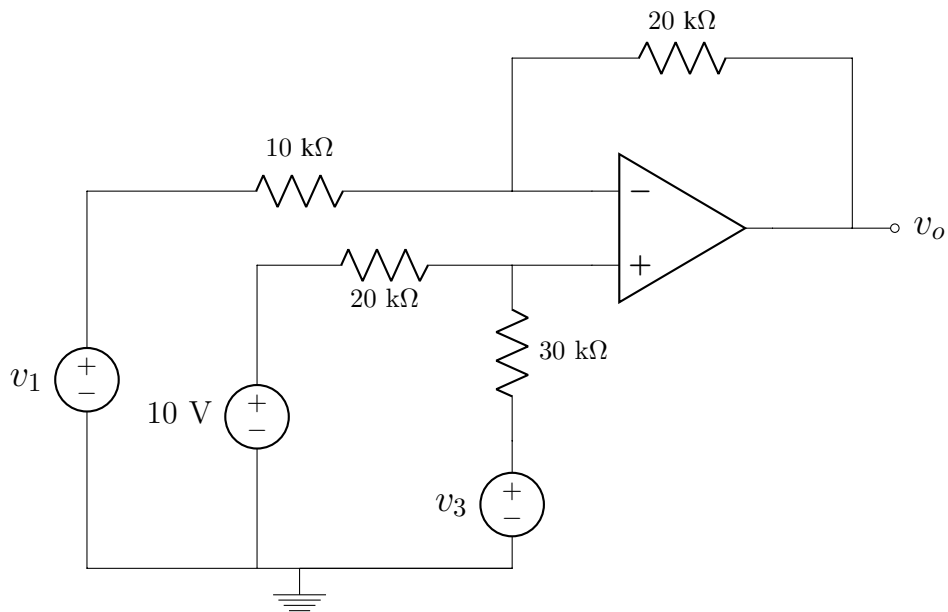
$$\begin{aligned}v_o &= \frac{-4}{2}v_s, \\-15 &\leq v_o \leq 15 \\-15 &\leq -2v_s \leq 15 \\-7.5 &\leq v_s \leq 7.5.\end{aligned}$$

Thus, the maximum value of $v_s = 7.5 \text{ V}$.

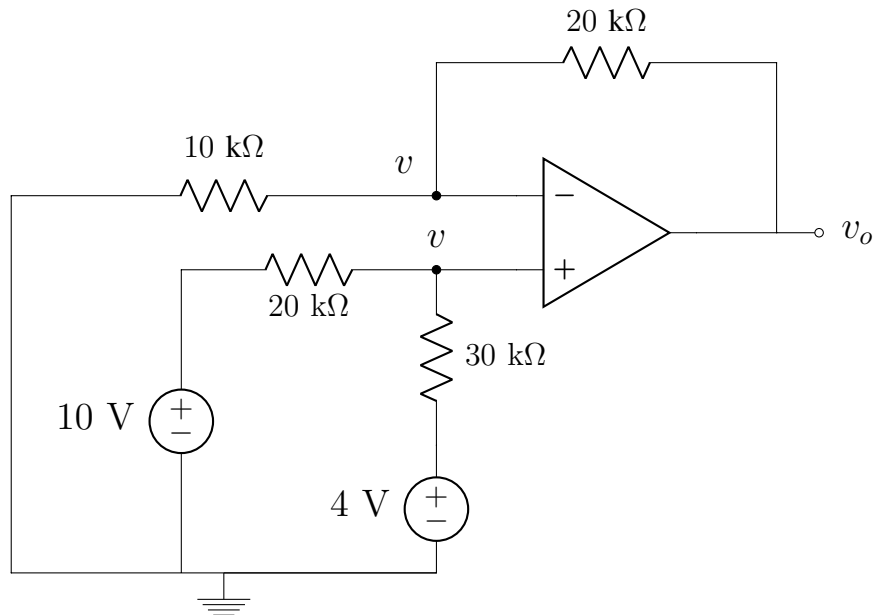
Questions 31–32 [4]

31 Find v_o if $v_1 = 0$ V and $v_3 = 4$ V.

32 Find v_1 if $v_o = 10$ V and $v_3 = 0$ V.



31:



Using KVL in the loop with v_2 and v_3 :

$$-10 + 20i + 30i + 4 = 0$$

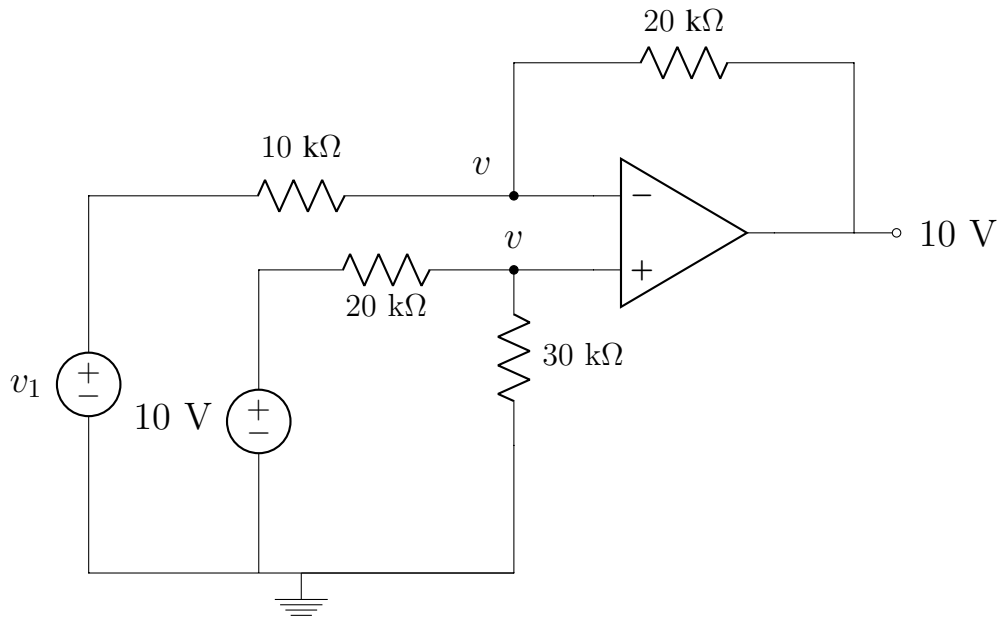
$$i = 0.12 \text{ mA.}$$

$$v = 4 + (0.12)(30) = 7.6 \text{ V.}$$

v becomes the input of a non-inverting configuration:

$$\begin{aligned} v_o &= v \left(1 + \frac{20}{10} \right) \\ &= 7.6 \left(1 + \frac{20}{10} \right) \\ v_o &= 22.8 \text{ V.} \end{aligned}$$

32:



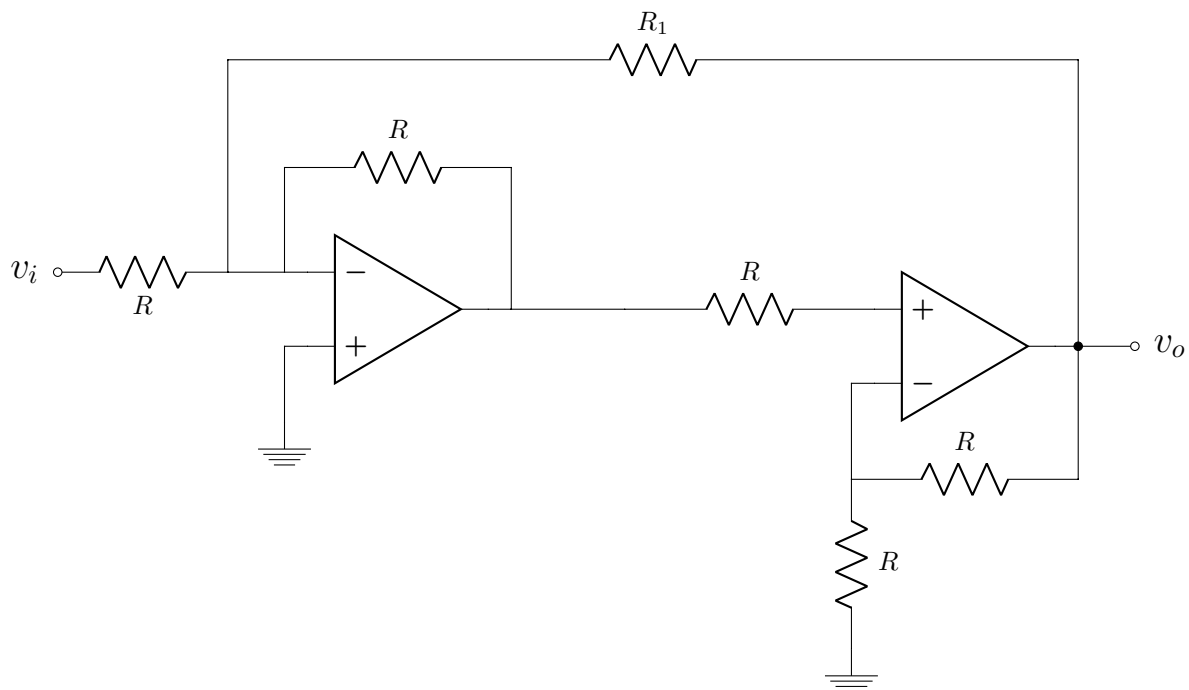
Recognising that the above circuit reduces to a difference amplifier:

$$\begin{aligned} 10 &= \left(\frac{10 + 20}{20 + 30} \right) \frac{30}{10}(10) - \frac{20}{10}(v_1) \\ v_1 &= 4 \text{ V} \end{aligned}$$

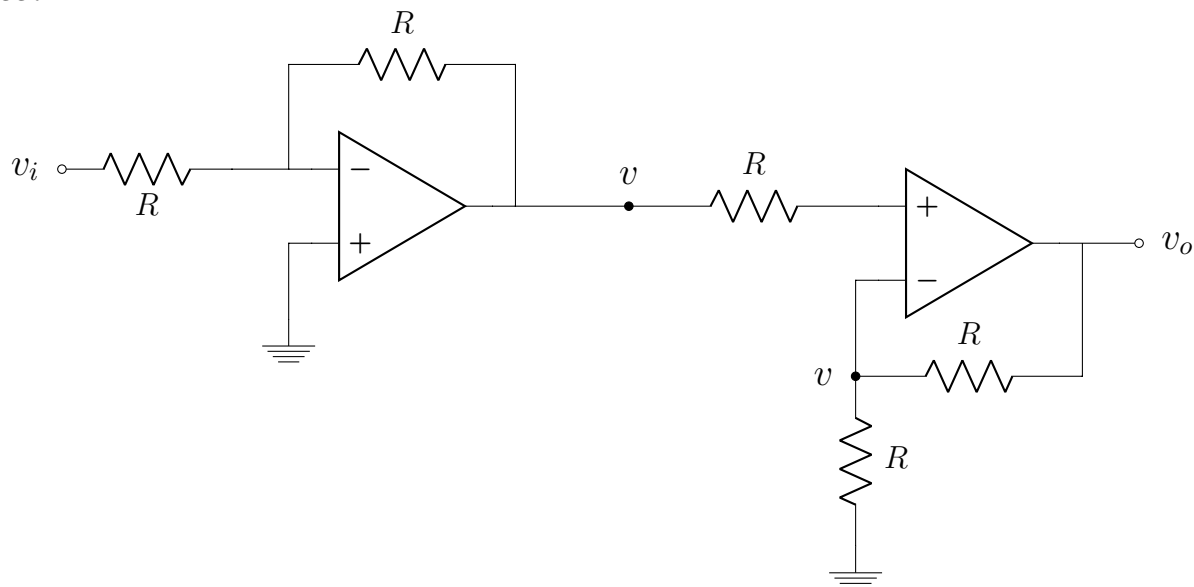
Questions 33–34 [4]

33 If $R_1 \rightarrow \infty$, then find $\frac{v_o}{v_i}$.

34 If $R_1 = R$ and $v_o = 10$ V, then find v_i .



33:



First stage – inverting amplifier:

$$v = -v_i$$

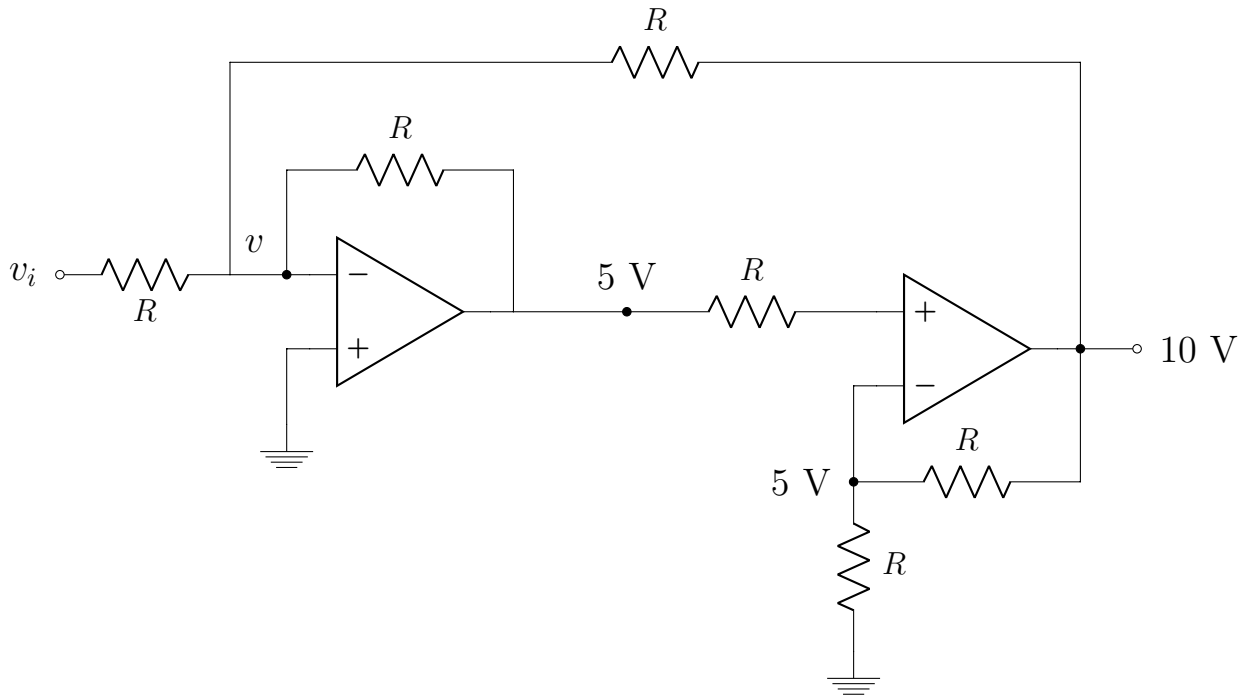
Second stage – non-inverting amplifier:

$$v_o = v \left(1 + \frac{R}{R} \right)$$

$$v_o = 2v.$$

Thus, $\frac{v_o}{v_i} = -2$.

34:



Using voltage division gives 5 V at the output node of the first op amp. Using KCL at node v :

$$\frac{0 - 10}{R} + \frac{0 - v_i}{R} + \frac{0 - 5}{R} = 0$$

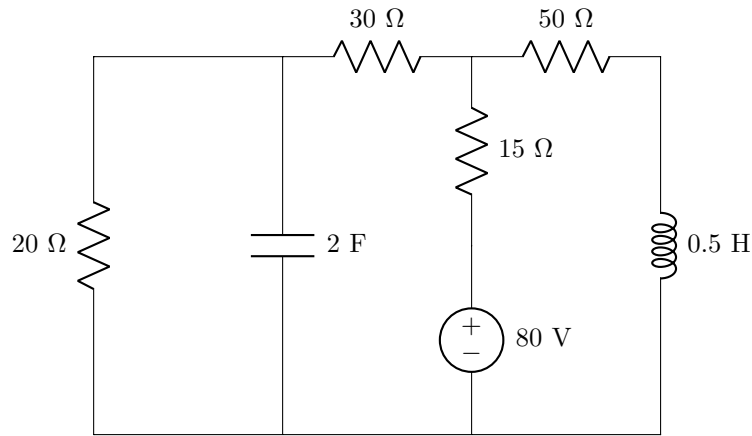
$$v_i = -15 \text{ V}.$$

Questions 35–36 [4]

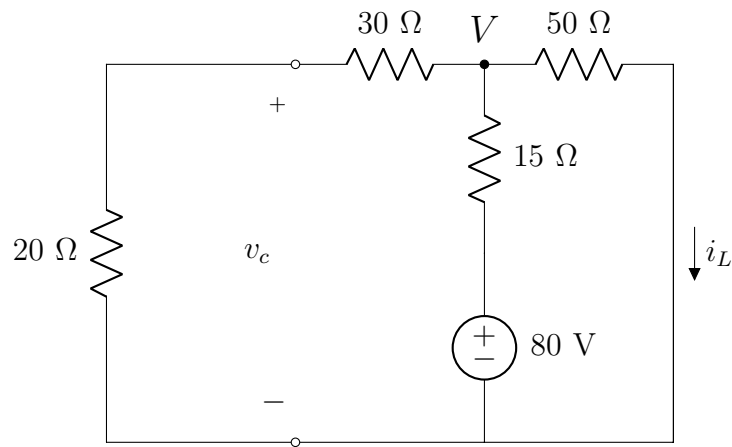
The circuit has reached a steady state.

35 Find the energy stored in the capacitor (E_C).

36 Find the energy stored in the inductor (E_L).



Solution:



KCL at node V :

$$\frac{V}{50} + \frac{V - 80}{15} + \frac{V}{50} = 0$$

$$V = 50 \text{ V.}$$

$v_c = 20 \text{ V}$ (voltage division at V) and $i_L = 1 \text{ A}$ (Ohm's law).

35:

$$E_c = \frac{1}{2} C v_c^2$$

$$= \frac{1}{2} (2) (20)^2$$

$$= 400 \text{ J.}$$

36:

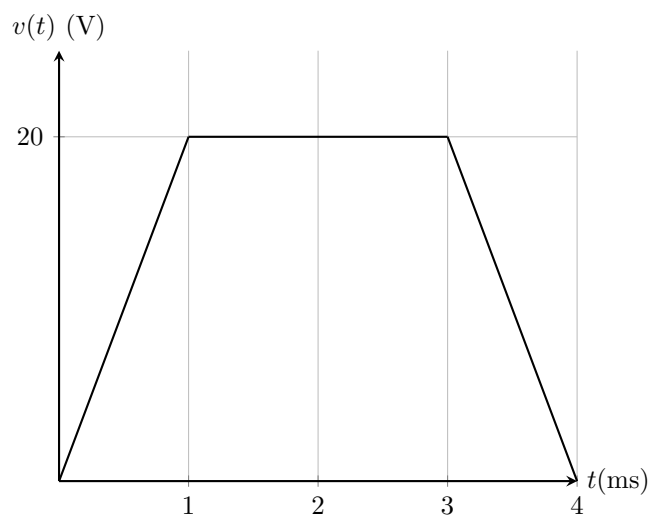
$$\begin{aligned}E_L &= \frac{1}{2}Li_L^2 \\&= \frac{1}{2}(0.5)(1)^2 \\&= 0.25 \text{ J.}\end{aligned}$$

Questions 37–38 [2]

The voltage across a 2 mF capacitor is shown below. Determine the current through the capacitor at $t = 0.5$ ms and $t = 2$ ms.

37 $i(0.5 \text{ ms})$

38 $i(2 \text{ ms})$



37:

$$\begin{aligned} i(0.5 \text{ ms}) &= C \frac{dv}{dt} \\ &= (0.002) \left(\frac{20}{10^{-3}} \right) \\ &= 40 \text{ A.} \end{aligned}$$

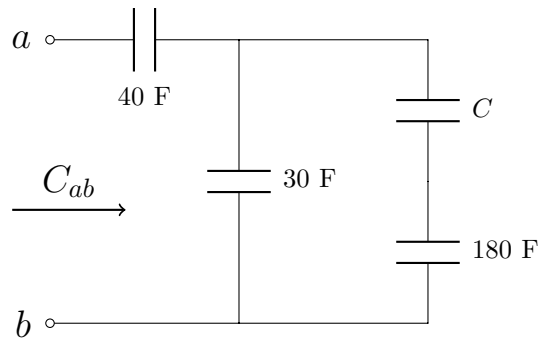
38:

$$i(2 \text{ ms}) = 0 \text{ A.}$$

Question 39 [2]

39

Find C if the equivalent capacitance w.r.t terminals $a - b$ (C_{ab}) is 30 F.



39:

Note that $a//b$ refers to $\frac{ab}{a+b}$.

$$C_{eq} = 30\text{ F} = 40//[30 + C//180]$$

Let $x = [30 + C//180]$:

$$30\text{ F} = 40//x$$

$$x = 120$$

$$120 = 30 + C//180$$

$$C = 180\text{ F}.$$

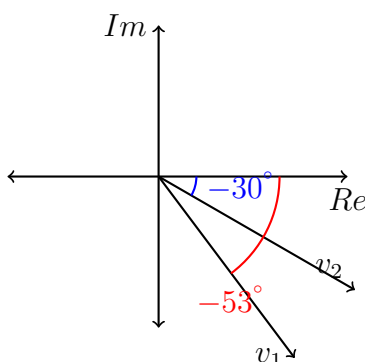
Questions 40–42 [4]		
Given two sinusoid signals $v_1(t) = 5 \cos(2t - 53^\circ),$ $v_2(t) = 12 \cos(2t - 30^\circ).$		
40	Choose the correct statement. 1) v_1 leads v_2. 2) v_2 leads v_1. Write 1 if statement 1) is correct or 2 if statement 2) is correct.	[1]
41	Find the phase angle θ between v_1 and v_2 (with $\theta > 0$).	[1]
42	Use phasors to find $\bar{V}_T = \bar{V}_1 - \bar{V}_2$ in polar form.	[2]

40:

$$\bar{V}_1 = 5 \angle -53^\circ.$$

$$\bar{V}_2 = 12 \angle -30^\circ.$$

As seen below, v_2 leads v_1 . Thus **statement 2** is true.



41: The phase difference θ is 23° .

42:

$$\begin{aligned}
 \bar{V}_T &= \bar{V}_1 - \bar{V}_2 \\
 &= 5 \angle -53^\circ - 12 \angle -30^\circ \\
 &= 7.66 \angle 165^\circ \text{ V.}
 \end{aligned}$$

Questions 43–44 [4]	
43	Simplify the following expression and give the answer in rectangular form (* refers to conjugation).
$\bar{V} = \left(\frac{10 + j20}{3 + j4} \right)^2 \sqrt{(10 + j5)^* (16 - j20)}$	
44	Find $\bar{I}_2$ in polar form.
$\begin{bmatrix} 20\angle -30^\circ & 2 - j3 \\ -5 & 3\angle 45^\circ \end{bmatrix} \begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5\angle 0^\circ \end{bmatrix}$	

43:

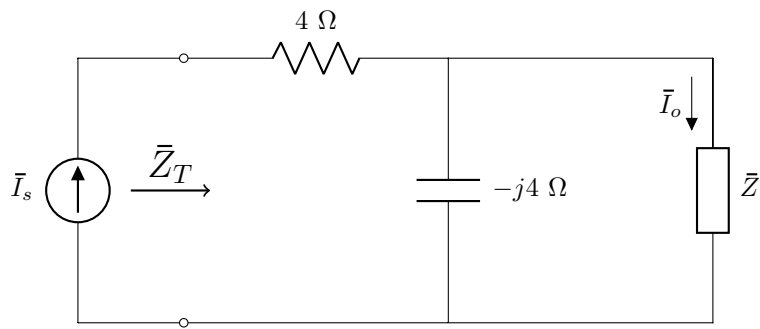
$$\begin{aligned}
 \bar{V} &= (18.72 + j7.04) [(10 - j5)(16 - j20)]^{\frac{1}{2}} \\
 &= (18.72 + j7.04) [16.92\angle -38.955^\circ] \\
 &= 321 - j107 \text{ V.}
 \end{aligned}$$

44:

$$\begin{aligned}
 \bar{I}_2 &= \frac{\begin{vmatrix} 20\angle -30^\circ & 0 \\ -5 & -5 \end{vmatrix}}{\begin{vmatrix} 20\angle -30^\circ & 2 - j3 \\ -5 & 3\angle 45^\circ \end{vmatrix}} \\
 &= 1.472\angle 149.55^\circ.
 \end{aligned}$$

Questions 45–46 [4]

- | | |
|-----------|--|
| 45 | Find $\bar{Z}$ in rectangular form if $\bar{Z}_T = 20 + j12 \Omega$. |
| 46 | If $\bar{I}_s = 4\angle 90^\circ$ A and $\bar{Z} = 4 - j4 \Omega$, find $\bar{I}_o$ in rectangular form. |

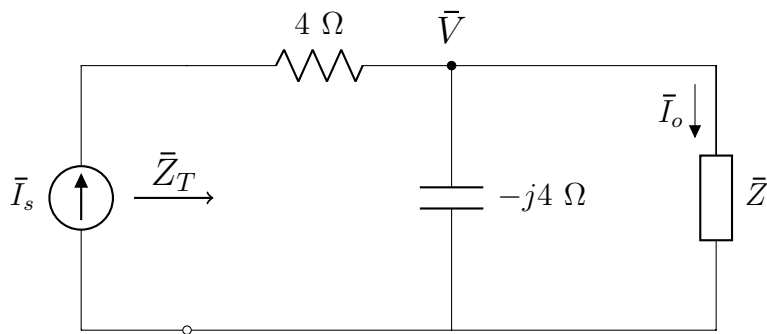


45:

$$\bar{Z}_T = 4 + \left[\frac{1}{-j4} + \frac{1}{\bar{Z}} \right]^{-1} = 20 + j12$$

$$\frac{1}{16 + j12} = \frac{1}{-j4} + \frac{1}{\bar{Z}}$$

$$\bar{Z} = 0.5 + j3.5 \Omega.$$



46:

$$\bar{I}_o = 4\angle 90^\circ \left(\frac{-j4}{-j4 + 4 - j4} \right)$$

$$= 0.8 + j1.6 \text{ A.}$$

Questions 47–50 [4]

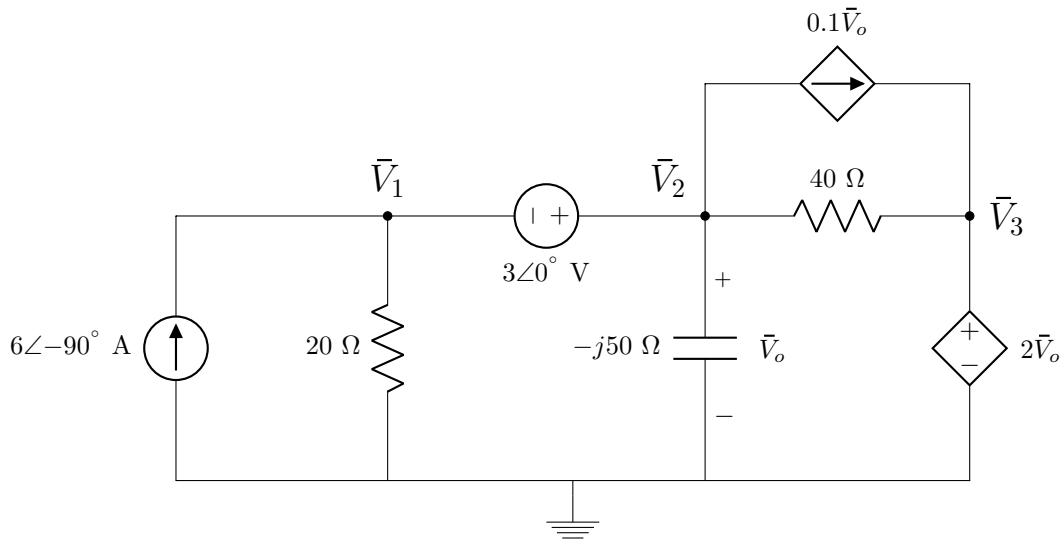
Use nodal analysis to set up two equations of the following form:

$$k\bar{V}_1 + l\bar{V}_2 = -3,$$

$$m\bar{V}_1 + (-4 + j15)\bar{V}_2 = n.$$

Find the complex values k , l , m , and n in **rectangular** form.

47	k	48	l
49	m	50	n



47-50:

Supernode:

$$\bar{V}_1 + 3 = \bar{V}_2$$

$$\bar{V}_1 - \bar{V}_2 = -3.$$

$$j6 + \frac{\bar{V}_1}{20} + 0.1\bar{V}_2 + \frac{\bar{V}_2 - 2\bar{V}_2}{40} + \frac{\bar{V}_2}{-j50} = 0$$

$$\bar{V}_1 \left[\frac{1}{20} \right] + \bar{V}_2 \left[0.1 - \frac{1}{40} + \frac{1}{-j50} \right] = -j6$$

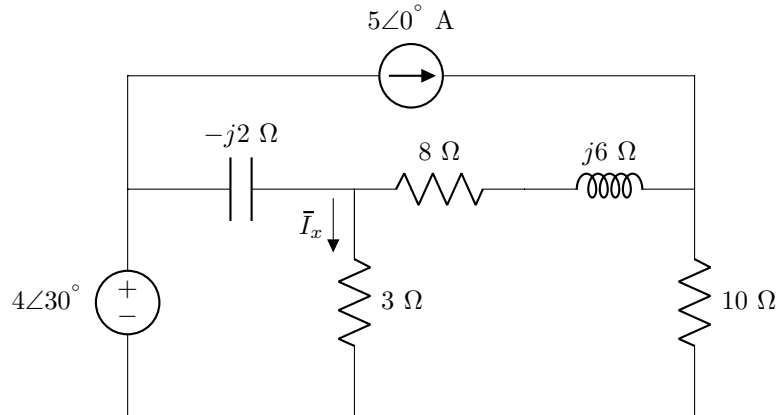
$$j10\bar{V}_1 + (-4 + j15)\bar{V}_2 = 1200.$$

Thus $k = 1$, $l = -1$, $m = j10$, and $n = 1200$.

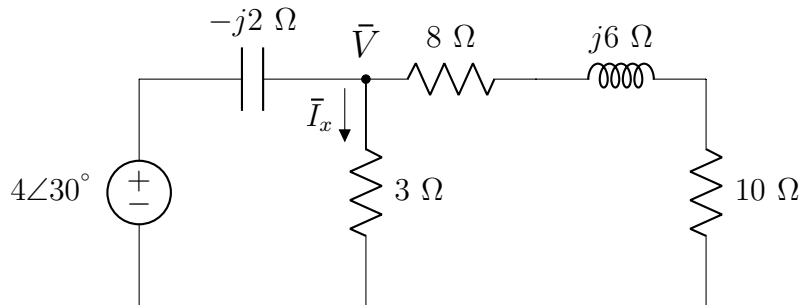
Question 51 [2]

51

Find $\bar{I}_x^V$, the contribution of the voltage source to $\bar{I}_x$, in **rectangular** form.



51:



KCL at node $\bar{V}$:

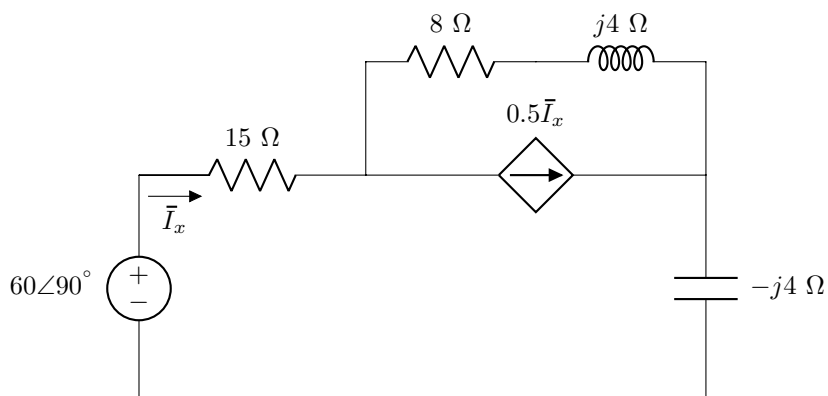
$$\begin{aligned} \frac{\bar{V}}{3} + \frac{\bar{V} - 4\angle 30^\circ}{-j2} + \frac{\bar{V}}{18 + j6} &= 0 \\ \bar{V} \left[\frac{1}{3} + \frac{1}{-j2} + \frac{1}{18 + j6} \right] &= \frac{4\angle 30^\circ}{-j2} \\ \bar{V} &= 3.24\angle 68.4^\circ \text{ V} \\ \bar{I}_x^V &= \frac{\bar{V}}{3} = 0.398 + j1.00 \text{ A.} \end{aligned}$$

Questions 52–54 [4]

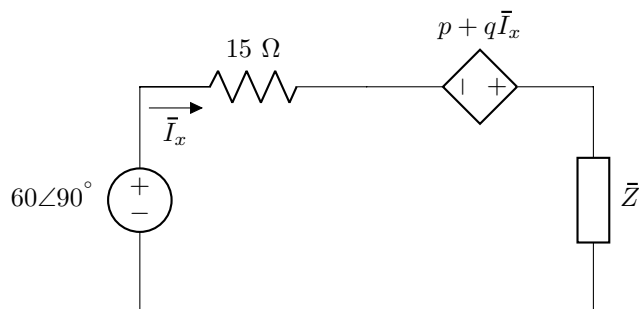
Use source transformation to determine p , q , and $\bar{Z}$ in *circuit B* (in **rectangular** form) such that it is equivalent to *circuit A*.

52	p	[1]
53	q	[1]
54	$\bar{Z}$	[2]

Circuit A:



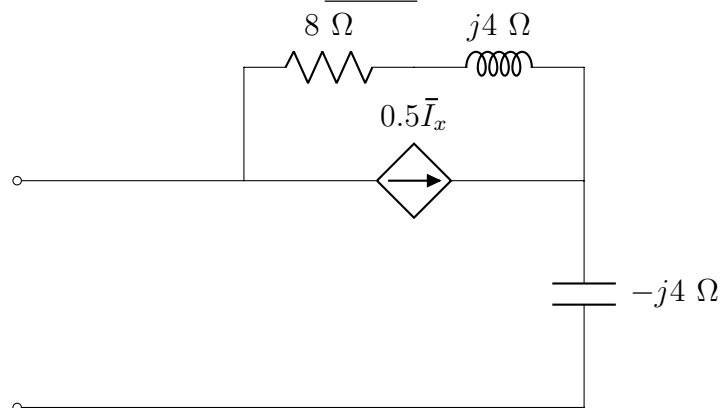
Circuit B:



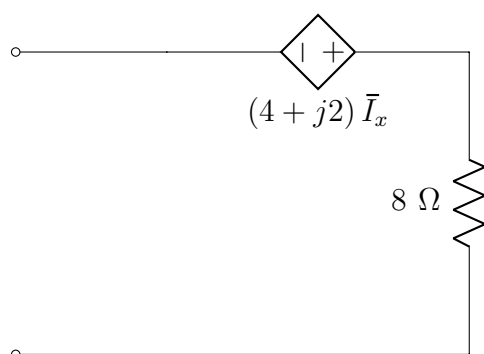
Turn over for additional space for Questions 52–54.

Additional space for Questions 52–54.

Step 0:



Step 1:



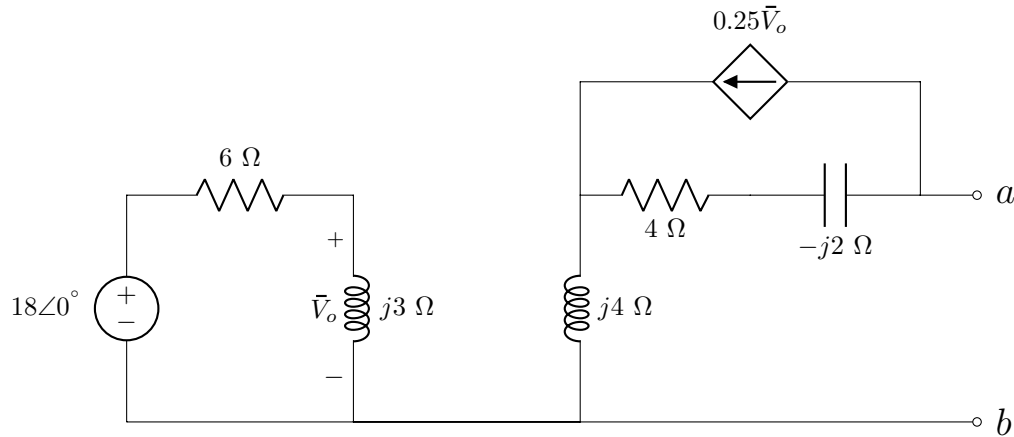
Thus, $p = 0$, $q = 4 + j2$, and $\bar{Z}_T = 8\ \Omega$.

Questions 55–56 [4]

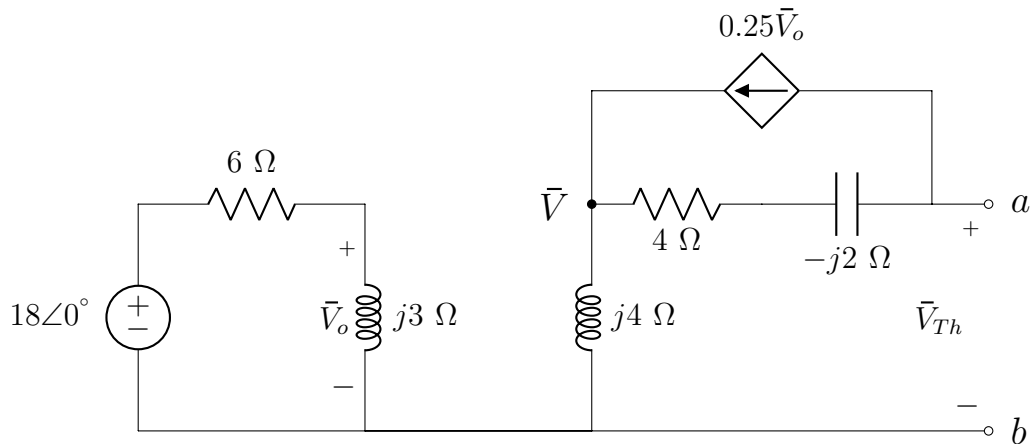
Find the Thevenin equivalent voltage $\bar{V}_{Th}$ and Norton equivalent current $\bar{I}_N$ w.r.t. terminals $a - b$ in **rectangular** form.

55 $\bar{V}_{Th}$

56 $\bar{I}_N$



55:



Voltage division for left hand part of the circuit. $\bar{V} = 0$ V because the current $0.25\bar{V}_o$ flows in a loop, pulling $\bar{V}$ to ground.

$$\bar{V}_o = 18 \left(\frac{j3}{6 + j3} \right) = 3.6 + j7.2 \text{ V}$$

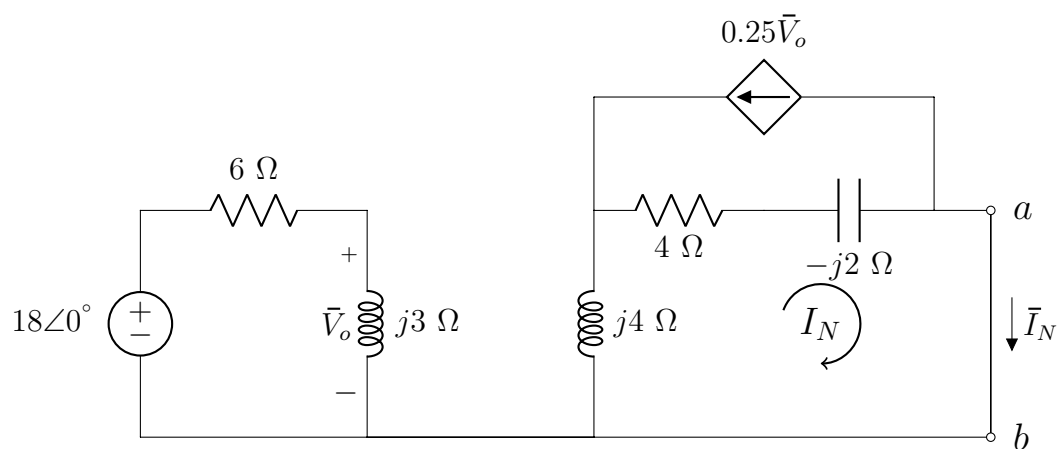
$$\bar{V} = 0.$$

Nodal analysis at $\bar{V}_{Th}$:

$$\frac{\bar{V}_{Th}}{4 - j2} + 0.25(3.6 + j7.2) = 0$$

$$\bar{V}_{Th} = 7.2 + j5.4 \text{ V.}$$

56:



Mesh analysis at $\bar{I}_N$:

$$j4\bar{I}_N + (4 - j2) [\bar{I}_N + 0.25(3.6 + j7.2)] = 0$$

$$\bar{I}_N [j4 + 4 - j2] = -(4 - j2)(0.25)(3.6 + j7.2)$$

$$\bar{I}_N = -1.98 - j0.36 \text{ A.}$$

PRACTICE PAGE

Assessment ID	2	0	1	9	E	B	N	1	1	1	E	0	1	
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Round off answers to 3 significant digits when applicable.

*All angles must be given in degrees in the range $[-180^\circ, 180^\circ]$. For complex numbers in polar form, unit of angle ($^\circ$) must be given in the **Answer/numeric** field. For **all other cases**, the unit of angle ($^\circ$) must be given in the **Unit/text** field.*

	Answer	Unit	Marks
SECTION 1			
01	$E = -300$	mJ	2
02	$Cost = 4320$	R	2
03	$P_I = -15$	W	2
04	$P_V = -80$	W	2
05	$R_{ab} = 20$	Ω	2
06	$R_{cd} = 8$	Ω	2
07	$R_2 = 19$	k Ω	2
08	$a = 1$	<i>No units required</i>	1
09	$b = -1$		1
10	$c = -3$		1
11	$d = -2$		1
12	$e = 1$		1
13	$f = -4$		1
14	$g = 35$		1
15	$h = -15$		1
16	$i = -8$		1
17	$j = 0.85$		1

	Answer	Unit	Marks
18	$I_1 = 15$	μA	2
19	$V_{CE} = 2.19$	V	2
20	$V_x^V = 10$	V	2
21	$V_x^I = 0$	V	2
22	$V_1 = -5$	V	2
23	$R_1 = 7.5$	Ω	2
24	$V_{Th} = 14$	V	2
25	$R_{Th} = 6$	Ω	2
26	$I_N = 7.5$	mA	2
27	$R = 400$	Ω	2
28	$I_N = 64$	mA	2
SECTION 2			
29	$v_s = 2.75$	V	2
30	$v_s = 7.5$	V	2
31	$v_o = 22.8$	V	2
32	$v_1 = 4$	V	2
33	$\frac{v_o}{v_1} = -2$	<i>No units required</i>	2
34	$v_i = -15$	V	2
35	$E_C = 400$	J	2
36	$E_L = 0.25$	J	2
37	$i(0.5 \text{ ms}) = 40$	A	1
38	$i(2 \text{ ms}) = 0$	A	1
39	$C = 180$	F	2

	Answer	Unit	Marks
40	<i>Correct Statement = 2</i>	<i>No units required</i>	1
41	$\theta = 23$	$^{\circ}$	1
42	$\bar{V}_T = 7.66 \angle 165^{\circ}$	<i>No units required</i>	2
43	$\bar{V} = 321 - j107$		2
44	$\bar{I}_2 = 1.47 \angle 150^{\circ}$		2
45	$\bar{Z} = 0.5 + j3.5$	Ω	2
46	$\bar{I}_o = 0.8 + j1.6$	A	2
47	$k = 1$	<i>No units required</i>	1
48	$l = -1$		1
49	$m = j10$		1
50	$n = 1200$		1
51	$\bar{I}_x^V = 0.398 + j1.00$	A	2
52	$p = 0$	<i>No units required</i>	1
53	$q = 4 + j2$		1
54	$\bar{Z}_T = 8$	Ω	2
55	$\bar{V}_{Th} = 7.2 + j5.4$	V	2
56	$\bar{I}_N = -1.98 - j0.36$	A	2